

4.3.10 Let H be the major arc mid-pt of $\widehat{BC}$, let HI intersects (ABC) again at G (it's well-known that G is the tangential point of the A -mixtilinear circle and the circumcircle). Let $P \neq A$ be the intersection of AY and (ABC) .
CLAIM 1 X, I, Y, D, G are concyclic, (and M the mid-pt of XY)
Proof.

Since $\angle XDY = \angle FDE = \angle FCE = \frac{C}{2} + \frac{B}{2} = 90^\circ - \frac{A}{2} = 180^\circ - \angle BIC = 180^\circ - \angle XIY$,
 X, I, Y, D are concyclic.

Meanwhile, $\angle FHE = \angle FAE = 180^\circ - \angle FDE = 90^\circ + \frac{A}{2}$

$$\angle BFH = 180^\circ - \angle BEH = 180^\circ - \angle BCH = 90^\circ + \frac{A}{2} = \angle FHE$$

$\therefore BFHE$ is an isosceles trapezoid and hence $FH \parallel BE$.

Now $\angle IGD = \angle HGD = \angle HFD = \angle IXD$, so I, X, G, D are concyclic, this completes the proof of the claim.

CLAIM 2 G, M, Y, P are concyclic.

Proof.

From CLAIM 1 and spiral similarity lemma it follows that G is the centre of spiral similarity sending XY to FE . Let L be the mid-pt of FE , from my proof of 4.3.8, IG is the I -symmedian of $\triangle BIC$ and hence the I -median of $\triangle FIE$ (since the two triangles are oppositely similar). Therefore L lies on FE , so $\triangle XMG \sim \triangle FLG$.

Meanwhile, since $AF = BF = HE$ (from $BFHE$ is isosceles trapezoid), $AFHE$ is

also an isosceles trapezoid.

Therefore $\angle AGF = \angle AEF = \angle HFE = \angle HGE$, together with GL is the G -median of $\triangle FGE$, GA is the G -symmedian of $\triangle FGE$. Therefore,

$$\triangle FGL \sim \triangle AGE, \triangle EGL \sim \triangle AGF.$$

Therefore, $\angle XMG = \angle FLG = \angle AEG = \angle APG = \angle YPG$, so G, M, Y, P are concyclic.

Now, we have $\angle MPA = \angle MPY = \angle MGY = \angle LGE = \angle HGE = \angle HFE = \angle AEF = \angle APF$
 $\therefore F, M, P$ are collinear as desired.

